MILITARY SPECIFICATION

CABLE ASSEMBLY W-1 OF AN/AQS-13

This amendment forms a part of Military Specification MIL-C-81452(AS) dated 15 November 1966 and has been approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 5

3.5.1.1: Delete in its entirety and substitute:

"3.5.1.1 Function - The Cable Assembly shall connect the hydraulic reeling machine slip rings to the transducer assembly and be capable of withstanding a 1000 pound tension. It shall be capable of being pulled over a four inch radius. The sonar cable including connector P7001A with adaptor J7001A (see 3.5.1.4.7) shall, when mated with the transducer, withstand without leaking and without change in performance a hydrostatic pressure of 500 psi. It shall be capable of withstanding, without damage or change in performance, the "crushing" loads encountered while passing over four inch radius sheaves at an average speed of 11 feet per second with a 150 pound load and being loaded on the reel and have a minimum breaking strength of 1500 pounds between the connector P7001A and the cable."

PAGE 5

3.5.1.2: Delete in its entirety and substitute:

"3.5.1.2 Form Factor - The cable length shall be 500 to 510 feet as shown on Figure 1. The diameter shall be not greater than 0.565 inch (except the connector assemblies) nor be less than 0.545 inch. The cable shall be not greater than 0.015 inch out-of-round and shall be straight within 0.015 inch over any four foot length when measured with a tensional load of 150 pounds. The cable shall not stretch by more than 0.2 percent of its total length with 270 pounds tensile load, and the cable diameter shall remain within the specified limits after stretching has occurred either during test or operation."

FSC 6150

AMSC N/A

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PAGE 6

3.5.1.4.1: Delete in its entirety and substitute:

"3.5.1.4.1 Electrical Conductors - The electrical conductors shall be as described in 3.6, Cable Construction."

PAGE 6

3.5.1.4.2: Delete in its entirety and substitute:

"3.5.1.4.2 Shielding Effectiveness - The fully terminated cable assembly shall have not less than 70 dB of isolation at 11 kHz between any randomly selected conductor within a bundle to any other randomly selected conductor in any other bundle. The cable shall be bundled such that this isolation exists between the 16 output signal leads and plus 30 volt lead to all other leads conducting AC signals. The connector pin assignments are specified as follows:

A	Xmit AC	a	temp gnd
B	Xmit AC	b	7L
C	Xmit AC	c	7R
D	BITE	d	8L
E	Compass AC	e	8R
F	1L	f	+30V
G	1R	g	temp 1
H	2L	h	temp 2
J	TVG	j	Sig Gnd
K	Compass AC	k	Depth max
L	2R	l	Spare
M	3L	m	Depth Min.
N	3R	n	xmit return AC
P	4L	p	xmit return AC
R	Compass 400 Hz	r	xmit return AC
S	Compass AC	s	xmit return AC
T	4R		
U	5L		
V	5R		
W	6L		
X	6R		
Z	Strength Member		

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3.5.1.4.3: Delete in its entirety:

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3.5.1.4.4: Delete in its entirety and substitute:

"3.5.1.4.4 Outer Jacket - The cable outer jacket shall be black polyethylene or polyurethane. It shall be capable of withstanding a hydrostatic pressure of 500 psig throughout the entire length without leaking. There shall be color coded markings every 50 feet along the entire length of the outer jacket as shown on Figure II."

PAGE 6

3.5.1.4.5: Delete in its entirety and substitute:

"3.5.1.4.5 Connectors - One end of the cable shall be terminated with a connector designated P7001A as described in 3.5.1.4.5.1 or an equal and shall be capable of withstanding a hydrostatic pressure of 500 psig without leaking. It shall be made of corrosion resistant material. The cable strength member shall be mechanically but not electrically connected to the shell of the connector. The other end of the cable shall be terminated with a screw type connector designated P5019B as described in 3.5.1.4.5.1 or an equal and having no positive mechanical connection to the cable except at the solder pots. The strength member and all other shields and wires shall be isolated electrically from the connector shell. In the event of total cable payout it shall be constructed to allow the cable to break away from its shell. The breakaway area of the cable shall be not larger than the diameter of the cable to allow the cable to continue to payout without causing depot level damage to the AN/AQS-13 Reeling Machine and Cable Wiper Assembly."

PAGE 6

3.5.1.4.5.1: Delete in its entirety and substitute:

"3.5.1.4.5.1 Cable Connectors - The connection to external circuits shall be provided as follows:

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Reference Designation	Receptacle or an Approved Equal	Function
P5019B	Screw type with a 28-21P insert. Bendix Electrical Products Part No. 65-522766-21P.	To connect the cable to the reeling machine slip ring assembly.
P7001A	A 28-21S insert Cannon Part No. 3106E28-21S-A69 F26-F42. This connector is further described in Figure III. Alternate Bendix Part No. 10-262453-21S.	To connect the cable to the sonar transducer via the adaptor.
J7001A	Adaptor	To connect P7001A to J7001."

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3.5.1.4.5.2: Delete in its entirety:

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3.5.1.6: Delete in its entirety and substitute:

"3.5.1.6 Insulation Resistance - All pins on a fully terminated cable assembly except n,p,r,s,j,A,B,C and z shall be greater than 500 megohms from each pin to all other pins when measured with 500 VDC megohmmeter. Pin z to J shall be not less than 50 megohms. Pin z to the housing of P7001A shall be not less than 50 megohms."

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3.5.1.7: Delete in its entirety and substitute:

"3.5.1.7 Conductor Resistance - All pins on a fully terminated cable assembly shall be not greater than 25 ohms pin to pin except for the following. Pin J shall be 1.0 ohm maximum. Pins A,B,C or n,p,r,s shall be not greater than 3.0 ohms. The series resistance of A,B,C and n,p,r,s shall be not greater than 4.0 ohms."

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3.5.1.9: Delete in its entirety and substitute:

"3.5.1.9 Reeling Endurance - The cable shall be capable of withstanding not less than 10,000 cycles, at each temperature point, consisting of not less than 90 degree bends over a conformal pulley 8 inches pitch diameter. The cycle tests shall be run at temperatures of -40°C, at ambient temperature, and at +55°C with a constant tensile load of 150 pounds on the cable."

PAGE 8

3.5.1.10: Delete in its entirety and substitute:

"3.5.1.10 Torsional Oscillation and Whirl - Using a device similar to that shown in Figure V of the specification, the cable and wet end termination shall be capable of withstanding 10,000 rotations without failure, with a load of 250 pounds, a torsional oscillation of plus and minus 180 degrees synchronous with a 360 degree rotation, and at ambient temperatures. The configuration of the test machine and cable lengths subjected to test shall be approved by the procuring activity prior to testing by the contractor."

PAGE 9

3.6: Delete in its entirety and substitute:

"3.6 Cable Construction - Materials and construction shall be in accordance with the requirements of MIL-C-915."

PAGE 9

3.6.1: Delete in its entirety and substitute:

"3.6.1 Center Strength Member - The center strength member shall have a breaking load greater than 1500 pounds and shall be insulated with natural low density polythene or an equal as approved by the procuring activity."

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3.6.2: Delete in its entirety and substitute:

"3.6.2 Conductors - All conductors, except those specified in 3.6.4 shall consist of unidirectionally stranded copper wire or equal. They shall be insulated with .006 inch nominal radial thickness high density polyethylene or equal as approved by the procuring activity. The conductors need not be color coded for identification purposes."

PAGE 9

3.6.3: Delete in its entirety and substitute:

"3.6.3 Conductor Bundles - Using the conductors described in 3.6.2 above, separate bundles shall be cabled. The length of lay of each bundle shall be chosen to minimize cross talk. Power and control leads shall be screened to prevent interference using copper wire."

PAGE 9

3.6.4: Delete in its entirety and substitute:

"3.6.4 Transmit Conductor(s) - The conductor(s) for the transmit lead(s) shall consist of unidirectionally stranded copper wire. The lead(s) shall be insulated with .015 inch nominal radial thickness high density polyethylene or equal. The lead(s) shall be unidirectionally wrapped with copper wire and lapped with mylar tape."

PAGE 16

Table I: Delete in its entirety and substitute the enclosed Table I.

PAGE 17

Figure I: All markings shall be .125 inch characters or larger.

PAGE 18

Figure II:

1. Delete the 21 ft. marking requirement in the marking Table and the Figure.

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2. Change the tolerance in Note 1 from ±1.0 to ±2.0 percent.

3. Change the tolerance of the 4 ft. 6.0 inch white band from ±5.0 inch to ±2.0 inch.

4. Change the tolerance of the 7 ft. 6 inch yellow band from ±1.0 inch to ±2.0 inch.

PAGE 20

Figure IV: Delete in its entirety.

PAGE 22

Figure VI: Delete in its entirety.

Table I. Preproduction Tests

Tests	Preprod	Initial Production	Sampling	Individual	Monthly
3.5.1.2 Visual and Dimensional	X	X	X	X	X
3.5.1.6 Insulation Resistance	X	X	X	X	X
3.5.1.7 Conductor Resistance	X	X	X	X	X
3.5.1.4.2 Crosstalk	X	X	X		
3.5.1.8 Hydrostatic Cable 1/	X	X	X	X	X
3.5.1.9 Reeling Endurance	X	X	X		
3.5.1.10 Flight Simulation	X	X	X		
3.5.1.5 Torque Balance	X	X	X		
3.5.1.11 Proof Load	X	X	X	X	X
3.5.1.12 Bottom Recovery	X	X	X		

1/Limited to only the hydrostatic pressure test of the wet end termination.

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Preparing Activity
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